End-to-End MLOps: Digit Image Classifier with DVC, GitHub Actions CI/CD, Docker and AWS

Project Report

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1. Introduction

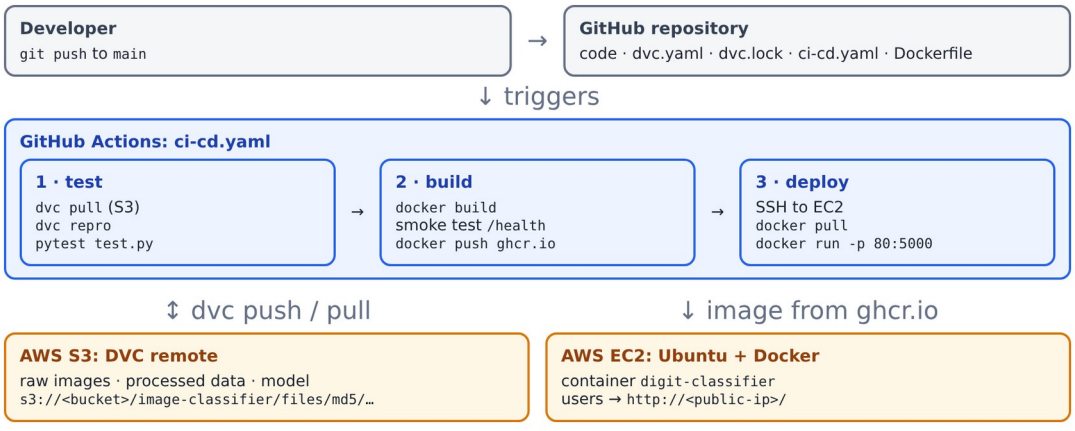
This project builds and deploys an **image classification application**, a handwritten-digit recognizer, using standard MLOps tooling:

Concern	Tool
Data and model versioning	**DVC**, with **AWS S3** as remote storage
Pipeline definition	`dvc.yaml` (prepare → train → evaluate)
Continuous integration and delivery	**GitHub Actions** (`.github/workflows/ci-cd.yaml`)
Containerization	**Docker** (`Dockerfile`), image published to GitHub Container Registry
Hosting	**AWS EC2**, the container served on port 80

Model. A Support Vector Machine (RBF kernel, `C = 10`, `gamma = 0.001`) with calibrated probabilities, trained on 1,797 grayscale 8×8 digit images (scikit-learn *digits* dataset, exported as PNG files).

Result. 99.17 % accuracy and 0.9917 macro-F1 on 360 held-out test images.

1.1 Architecture



System architecture

1.2 Repository layout

File	Purpose
`.github/workflows/ci-cd.yaml`	CI/CD workflow
`dvc.yaml`, `dvc.lock`, `params.yaml`	DVC pipeline, lock file, hyper-parameters
`data/raw.dvc`	DVC pointer for the 1,797-image dataset
`export_dataset.py`	exports the dataset as PNGs to `data/raw/<label>/`
`prepare.py`, `train.py`, `evaluate.py`	pipeline stages
`model.py`	preprocessing, model definition, prediction
`app.py`, `templates/index.html`	Flask web application
`test.py`	12 automated tests
`Dockerfile`, `.dockerignore`	container image
`deploy/ec2-user-data.sh`	EC2 bootstrap (installs Docker)
`deploy/deploy-manual.sh`	manual deployment script
`requirements.txt`, `requirements-dev.txt`	runtime and development dependencies

2. Step 1: Application

`model.py` holds the ML logic:

- `preprocess(image)` converts **any** uploaded image to the 64-value feature vector the model expects. It converts to grayscale, resizes to 8×8, inverts dark-on-light images (the training digits are light-on-dark) and scales the pixels to 0–16.
- `build_model()` creates the calibrated SVM. `predict()` returns `{"label", "confidence"}`.

`app.py` is a Flask application, served by **gunicorn** inside Docker:

Method	Route	Purpose
GET	`/`	Web page: draw a digit on a canvas, or upload an image
GET	`/health`	Health check, used by Docker `HEALTHCHECK`, CI smoke test and deploy verification
POST	`/predict`	Form field `image` → JSON prediction; returns 400 for a missing or invalid file

3. Step 2: DVC pipeline

3.1 Initialise DVC and version the raw images

```
dvc init
python export_dataset.py        # 1,797 PNGs -> data/raw/0 ... data/raw/9
dvc add data/raw                # creates data/raw.dvc; data/raw is git-ignored
```

```
Terminal — 01_dvc_init

user@ml:~/image-classifier-ci$ dvc --version
3.67.1

user@ml:~/image-classifier-ci$ dvc init
Initialized DVC repository.

You can now commit the changes to git.

+---------------------------------------------------------------------+
|                                                                     |
|        DVC has enabled anonymous aggregate usage analytics.         |
|     Read the analytics documentation (and how to opt-out) here:     |
|             <https://dvc.org/doc/user-guide/analytics>              |
|                                                                     |
+---------------------------------------------------------------------+

What's next?
------------
- Check out the documentation: <https://dvc.org/doc>
- Get help and share ideas: <https://dvc.org/chat>
- Star us on GitHub: <https://github.com/treeverse/dvc>

user@ml:~/image-classifier-ci$ git add .dvc .dvcignore && git commit -qm 'Initialize DVC' && git log --oneline -1
280c287 Initialize DVC
```

DVC initialised inside the Git repository

```
Terminal — 02_dvc_add_raw_images

user@ml:~/image-classifier-ci$ python export_dataset.py
Exported 1797 images to data/raw/

user@ml:~/image-classifier-ci$ ls data/raw && ls data/raw/7 | head -5
0
1
2
3
4
5
6
7
8
9
0007.png
0017.png
0027.png
0043.png
0044.png

user@ml:~/image-classifier-ci$ dvc add data/raw

To track the changes with git, run:

        git add data/raw.dvc data/.gitignore

To enable auto staging, run:

        dvc config core.autostage true

user@ml:~/image-classifier-ci$ cat data/raw.dvc
outs:
- md5: ee71162fd92c27e7d2903db86174f61f.dir
  size: 340893
  nfiles: 1797
  hash: md5
  path: raw
```

Raw image dataset exported and tracked with `dvc add`

3.2 Pipeline definition (`dvc.yaml`)

```yaml
stages:
  prepare:
    cmd: python prepare.py
    deps: [prepare.py, model.py, data/raw]
```

```
    params: [prepare]
    outs: [data/processed]
  train:
    cmd: python train.py
    deps: [train.py, model.py, data/processed]
    params: [train]
    outs: [models/digit_classifier.joblib]
  evaluate:
    cmd: python evaluate.py
    deps: [evaluate.py, models/digit_classifier.joblib, data/processed]
    metrics:
      - metrics.json: {cache: false}
```

Stage	Input	Output
prepare	`data/raw/*/*.png`, `params.prepare`	`data/processed/{train,test}.npz` (stratified 80/20 split)
train	training split, `params.train`	`models/digit_classifier.joblib`
evaluate	model and test split	`metrics.json`

`dvc repro` runs only the stages whose dependencies changed. `dvc.lock` records the MD5 hash of every input and output, so each Git commit pins an exact dataset and model version.

```
Terminal — 04_dvc_repro

user@ml:~/image-classifier-ci$ cat params.yaml
prepare:
  test_size: 0.2
  random_state: 42

train:
  C: 10
  gamma: 0.001

user@ml:~/image-classifier-ci$ dvc repro
'data/raw.dvc' didn't change, skipping
Running stage 'prepare':
> python prepare.py
1797 images -> train: 1437, test: 360
Generating lock file 'dvc.lock'
Updating lock file 'dvc.lock'

Running stage 'train':
> python train.py
Trained on 1437 images, model saved to /home/user/image-classifier-ci/models/digit_classifier.joblib
Updating lock file 'dvc.lock'

Running stage 'evaluate':
> python evaluate.py
{'accuracy': 0.9917, 'f1_macro': 0.9917, 'test_images': 360}
Updating lock file 'dvc.lock'

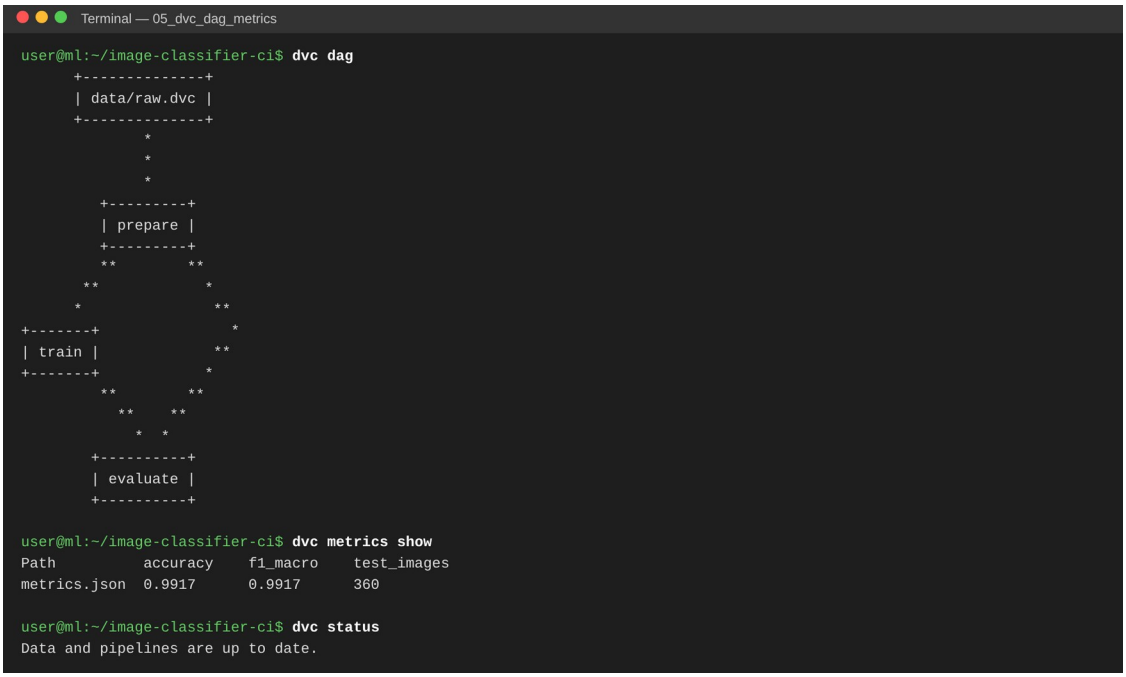
To track the changes with git, run:

        git add dvc.lock data/.gitignore models/.gitignore

To enable auto staging, run:

        dvc config core.autostage true
Use `dvc push` to send your updates to remote storage.
```

`dvc repro` running all three stages



Pipeline DAG and metrics

4. Step 3: AWS S3 as DVC remote storage

4.1 IAM user and AWS CLI

1. In **IAM** → **Users** → **Create user**, create `dvc-ci-user` with the policy `AmazonS3FullAccess` (in production, a policy limited to the single bucket).
2. Under **Security credentials** → **Create access key**, choose the CLI use case.
3. Run `aws configure` with that key and region `us-east-1`.

[INSERT SCREENSHOT] `aws configure list` and `aws sts get-caller-identity`

4.2 Bucket and DVC remote

```
aws s3 mb s3://<bucket-name> --region us-east-1
dvc remote add -d s3remote s3://<bucket-name>/image-classifier
dvc remote modify s3remote region us-east-1
dvc push
```

```
Terminal — 03_dvc_remote_s3
user@ml:~/image-classifier-ci$ dvc remote add -f -d s3remote s3://YOUR-BUCKET-NAME/image-classifier
Setting 's3remote' as a default remote.

user@ml:~/image-classifier-ci$ dvc remote modify s3remote region us-east-1

user@ml:~/image-classifier-ci$ dvc remote list
s3remote        s3://YOUR-BUCKET-NAME/image-classifier  (default)

user@ml:~/image-classifier-ci$ cat .dvc/config
[core]
    remote = s3remote
['remote "s3remote"']
    url = s3://YOUR-BUCKET-NAME/image-classifier
    region = us-east-1
```

S3 configured as the default DVC remote

[INSERT SCREENSHOT] `dvc push` output, showing the files uploaded to S3

[INSERT SCREENSHOT] S3 console: bucket → `image-classifier/files/md5/…` objects

DVC stores objects under their MD5 hash, so the object keys in S3 match the hashes in `dvc.lock` and `data/raw.dvc`. On a new machine, or in CI, `dvc pull` restores the exact dataset and model for the checked-out commit.

4.3 Verification against a local S3 emulator

The S3 workflow was also tested end to end against **moto**, a local server that implements the S3 API, before AWS credentials were available. The `aws` and `dvc` commands are the same as for real S3, except for the `--endpoint-url` / `endpointurl` setting. Every screenshot in this subsection is labeled *LOCAL S3 EMULATOR (moto), not real AWS*.

```
Terminal — 12_s3emu_aws_cli [LOCAL S3 EMULATOR (moto) — not real AWS]
user@ml:~/image-classifier-ci$ aws --version
aws-cli/1.46.2 Python/3.13.17 Linux/6.18.44-fc-v153 botocore/1.43.62

user@ml:~/image-classifier-ci$ aws configure list
      Name                    Value             Type    Location
      ----                    -----             ----    --------
   profile                <not set>             None    None
access_key     ****************ting             env
secret_key     ****************ting             env
    region                us-east-1              env    AWS_DEFAULT_REGION

user@ml:~/image-classifier-ci$ aws --endpoint-url http://localhost:5050 sts get-caller-identity
{
    "UserId": "AKIAIOSFODNN7EXAMPLE",
    "Account": "123456789012",
    "Arn": "arn:aws:sts::123456789012:user/moto"
}
```

AWS CLI configured (emulator credentials)

```
Terminal — 13_s3emu_create_bucket [LOCAL S3 EMULATOR (moto) — not real AWS]
user@ml:~/image-classifier-ci$ aws --endpoint-url http://localhost:5050 s3 mb s3://image-classifier-dvc-demo
make_bucket: image-classifier-dvc-demo

user@ml:~/image-classifier-ci$ aws --endpoint-url http://localhost:5050 s3 ls
2026-10-25 18:45:21 image-classifier-dvc-demo
```

Bucket created with `aws s3 mb`

```
Terminal — 14_s3emu_dvc_remote [LOCAL S3 EMULATOR (moto) — not real AWS]

user@ml:~/image-classifier-ci$ dvc remote add --local s3emu s3://image-classifier-dvc-demo/image-classifier

user@ml:~/image-classifier-ci$ dvc remote modify --local s3emu endpointurl http://localhost:5050

user@ml:~/image-classifier-ci$ dvc remote list
s3remote        s3://YOUR-BUCKET-NAME/image-classifier  (default)
s3emu   s3://image-classifier-dvc-demo/image-classifier
```

S3 remote added for DVC

```
Terminal — 15_s3emu_dvc_push [LOCAL S3 EMULATOR (moto) — not real AWS]

user@ml:~/image-classifier-ci$ dvc push -r s3emu
1802 files pushed

user@ml:~/image-classifier-ci$ dvc status -r s3emu
Cache and remote 's3emu' are in sync.
```

`dvc push`: 1,802 files uploaded (1,797 images, processed data, model)

```
Terminal — 16_s3emu_bucket_contents [LOCAL S3 EMULATOR (moto) — not real AWS]

user@ml:~/image-classifier-ci$ aws --endpoint-url http://localhost:5050 s3 ls s3://image-classifier-dvc-demo/image-
classifier/ --recursive --summarize | tail -3

Total Objects: 1802
   Total Size: 1050514

user@ml:~/image-classifier-ci$ grep -A3 'path: models/digit_classifier.joblib' dvc.lock | tail -3
      hash: md5
      md5: 2f8f42f6b5b6396aee95579c40811486
      size: 415252

user@ml:~/image-classifier-ci$ aws --endpoint-url http://localhost:5050 s3 ls s3://image-classifier-dvc-demo/image-
classifier/files/md5/2f/8f42f6b5b6396aee95579c40811486 --human-readable
2026-10-25 18:45:36  405.5 KiB 8f42f6b5b6396aee95579c40811486
```

Bucket contents: the model object key matches its MD5 hash in `dvc.lock`

```
Terminal — 17_s3emu_dvc_pull_restore [LOCAL S3 EMULATOR (moto) — not real AWS]

user@ml:~/image-classifier-ci$ ls data models
data:
raw.dvc

models:

user@ml:~/image-classifier-ci$ dvc pull -r s3emu
A       data/processed/
A       data/raw/
A       models/digit_classifier.joblib
1802 files fetched and 1800 files added

user@ml:~/image-classifier-ci$ ls data && ls data/raw/3 | wc -l && ls -l models/digit_classifier.joblib
processed
raw
raw.dvc
183
-rw-r--r-- 1 root root 415252 Oct 25 18:46 models/digit_classifier.joblib

user@ml:~/image-classifier-ci$ dvc status
Data and pipelines are up to date.
```

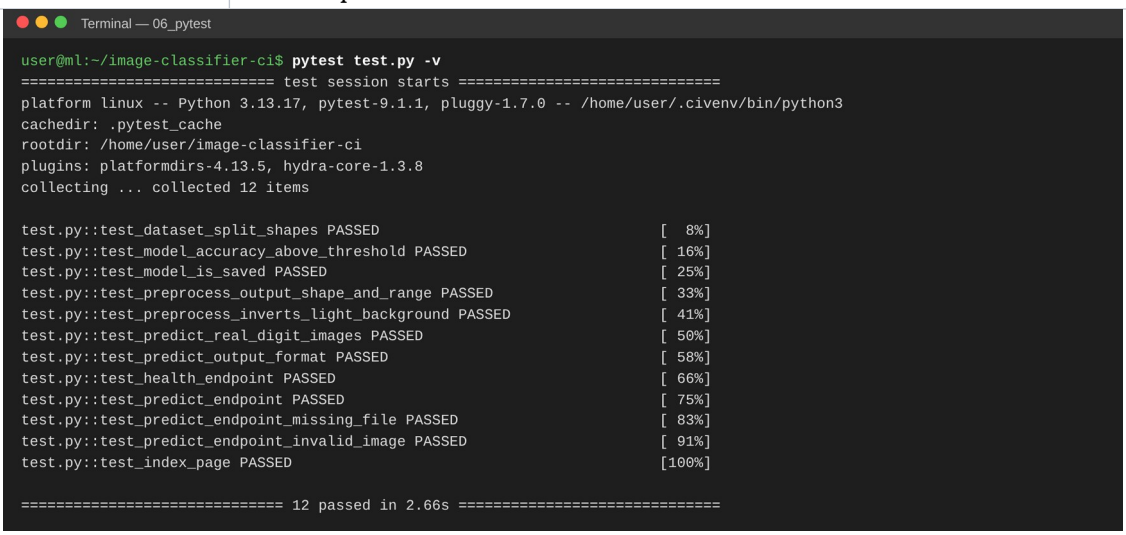
Restore test: local data, model and cache deleted, then recovered with `dvc pull`

The restore test confirms that the S3 remote holds everything needed to rebuild the project state. After deleting `data/raw`, `data/processed`, the model and the DVC cache, `dvc pull` fetched all 1,802 files and `dvc status` reported the workspace up to date.

5. Step 4: Automated tests (`test.py`)

`test.py` contains 12 pytest tests:

Area	What is verified
Data	split shapes, all 10 classes present
Model	test accuracy ≥ 0.95, model file is written
Preprocessing	any image → 1×64 vector in [0, 16]; light backgrounds are inverted
Prediction	≥ 45 of 50 real digit images correct; output format
API	`/` renders, `/health` returns 200, `/predict` returns the correct label, 400 on missing or invalid upload



All 12 tests passing

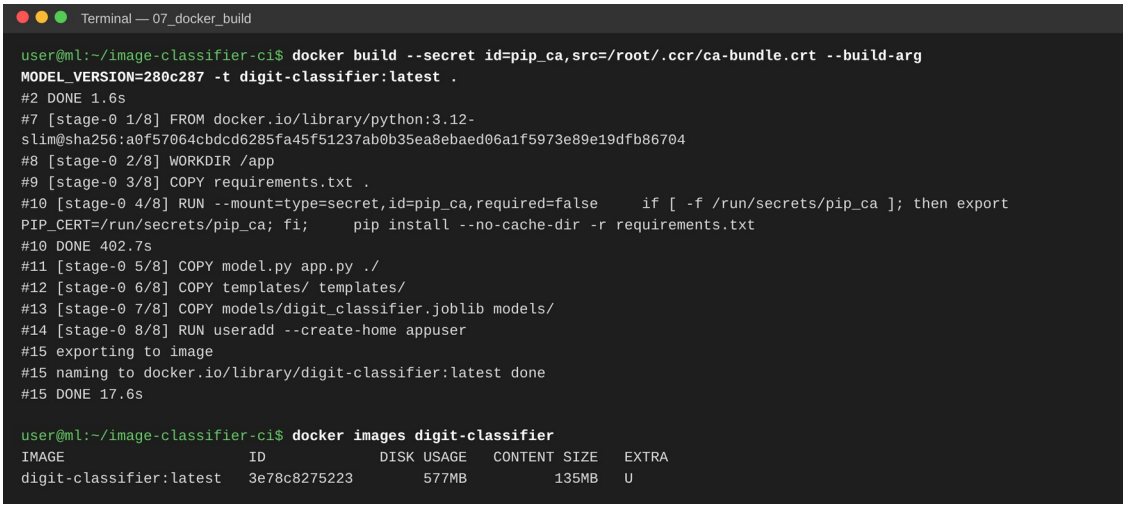
6. Step 5: Docker

```dockerfile
FROM python:3.12-slim
WORKDIR /app
COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt   # (+ optional proxy-CA secret mount)
COPY model.py app.py ./
COPY templates/ templates/
COPY models/digit_classifier.joblib models/          # DVC-tracked model
ARG MODEL_VERSION=dev
ENV MODEL_VERSION=${MODEL_VERSION}
RUN useradd --create-home appuser
USER appuser
EXPOSE 5000
HEALTHCHECK CMD python -c "import urllib.request;
urllib.request.urlopen('http://localhost:5000/health')"
CMD ["sh", "-c", "gunicorn --bind 0.0.0.0:${PORT} --workers 2 app:app"]
```

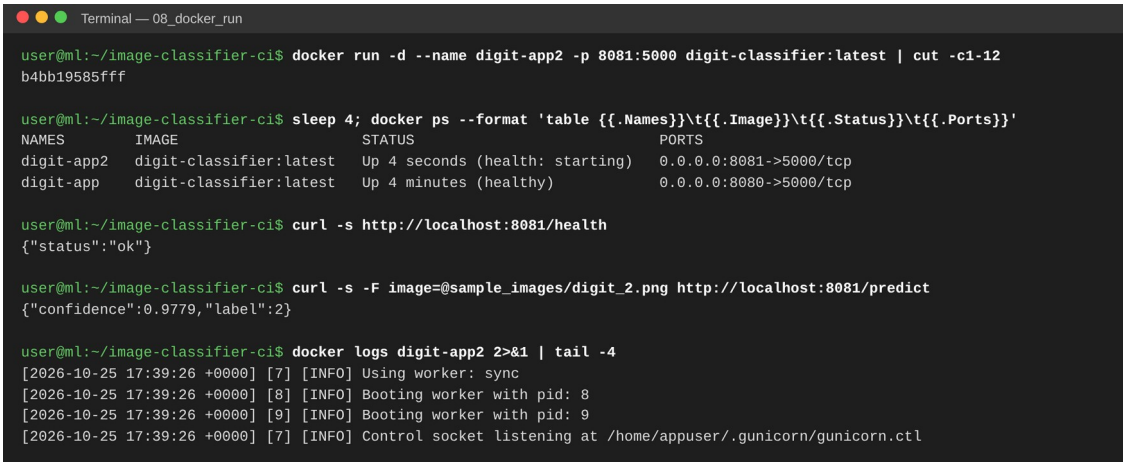
Design choices:

- A **slim base image** and only the runtime requirements. DVC and pytest stay out of the image.
- The container runs as a **non-root user**.
- The **model is copied in at build time** from the DVC output, so every image tag maps to exactly one model version. The Git SHA is passed in as `MODEL_VERSION` and shown in the UI footer.

- `.dockerignore` excludes the raw dataset, `.git` and the DVC cache.



Docker image build



Running the container and calling the API

7. Step 6: GitHub Actions CI/CD (`ci-cd.yaml`)

The workflow runs on every push and pull request to `main`/`master`, and can also be started by hand (`workflow_dispatch`). It has three jobs:

1. `test` (CI)

- Checks out the code, sets up Python 3.12 and installs `requirements-dev.txt`.
- If `AWS_ACCESS_KEY_ID` is configured, it logs in to AWS and runs **`dvc pull`** from S3. Otherwise it regenerates the dataset.
- Runs **`dvc repro`** and `dvc metrics show`, then **`pytest test.py`**.
- Uploads the trained model and the test results (JUnit XML, `metrics.json`) as artifacts.

2. `build` (CD, not run for pull requests)

- Downloads the model artifact and builds the Docker image, tagged `latest` and with the commit SHA.

- **Smoke-tests** the image: starts the container, waits for `/health`, sends a real prediction.
- Pushes the image to **GitHub Container Registry**: `ghcr.io/sidhantn12/image-classifier-ci`.

3. `deploy` (CD, runs once the `EC2_HOST` variable is set)

- SSHes into EC2 (`appleboy/ssh-action`), logs in to GHCR with the job token, pulls the new image, replaces the running container (`--restart unless-stopped -p 80:5000`) and prunes old images.
- Verifies the deployment by polling `http://<EC2_HOST>/health` from the runner.

Required repository settings:

Type	Name	Purpose
Secret	`AWS_ACCESS_KEY_ID`, `AWS_SECRET_ACCESS_KEY`	`dvc pull` from S3
Secret	`EC2_SSH_KEY`	private key of the EC2 key pair
Variable	`EC2_HOST`	public IP of the instance (enables the deploy job)
Variable	`EC2_USER`	`ubuntu`

Run #1 of the workflow (commit `7a7988b`) completed successfully. The `test` and `build` jobs passed and the image was pushed to GHCR. The `deploy` job was skipped because `EC2_HOST` was not yet configured.

```
Terminal — 09_github_actions_cicd_run

user@ml:~/image-classifier-ci$ curl -s https://api.github.com/repos/SidhantN12/image-classifier-ci/actions/runs/40217311681
(summarised)
Workflow : CI/CD  (#1)
Trigger  : push to main  @ 7a7988b  "Add DVC pipeline, Docker image, web UI and CI/CD workflow wi"
Status   : completed / SUCCESS
Started  : 2026-10-25T17:36:52Z   Finished: 2026-10-25T17:38:45Z
URL      : https://github.com/SidhantN12/image-classifier-ci/actions/runs/40217311681

✔ JOB  Test (DVC pipeline + pytest)  [success]
     ✔ Check out code
     ✔ Set up Python
     ✔ Install dependencies
     – Configure AWS credentials
     – Pull data and model from S3 (DVC)
     ✔ Regenerate raw dataset (no S3 credentials configured)
     ✔ Reproduce DVC pipeline
     ✔ Run tests
     ✔ Upload model
     ✔ Upload test results

✔ JOB  Build & push Docker image  [success]
     ✔ Check out code
     ✔ Download model
     ✔ Image name
     ✔ Log in to GitHub Container Registry
     ✔ Build image
     ✔ Smoke-test container
     ✔ Push image

– JOB  Deploy to AWS EC2  [skipped]

Artifacts:
   test-results        693 bytes
   model            105111 bytes

Image pushed: ghcr.io/sidhantn12/image-classifier-ci:latest, :7a7988b...
```

CI/CD run #1, from the GitHub API

> **[INSERT SCREENSHOT]** GitHub → Actions → CI/CD run: workflow graph with all three jobs green

[INSERT SCREENSHOT] Expanded "Run tests" step showing `12 passed`

[INSERT SCREENSHOT] "Deploy to AWS EC2" job log

8. Step 7: AWS EC2 deployment

8.1 Launch the instance

1. **EC2 → Launch instance**: *Ubuntu Server 24.04 LTS*, `t2.micro` or `t3.micro` (free tier).
2. **Key pair**: create `image-classifier-key` (.pem). Its content goes in the GitHub secret `EC2_SSH_KEY`.
3. **Security group** inbound rules: SSH (22) from *My IP*; HTTP (80) from `0.0.0.0/0`.
4. **Advanced details → User data**: the contents of `deploy/ec2-user-data.sh`, which installs and enables Docker and adds `ubuntu` to the `docker` group.
5. Launch the instance and copy its **Public IPv4 address** into the GitHub variable `EC2_HOST`.

[INSERT SCREENSHOT] EC2 instance list: instance *Running*, public IPv4 visible

[INSERT SCREENSHOT] Security group inbound rules (22, 80)

8.2 Deploy

Re-run the workflow (**Actions → CI/CD → Run workflow**) or push a commit. The `deploy` job then pulls the image on the instance and starts it. To deploy by hand instead:

```
ssh -i image-classifier-key.pem ubuntu@<EC2-IP>
./deploy-manual.sh ghcr.io/sidhantn12/image-classifier-ci:latest
docker ps
```

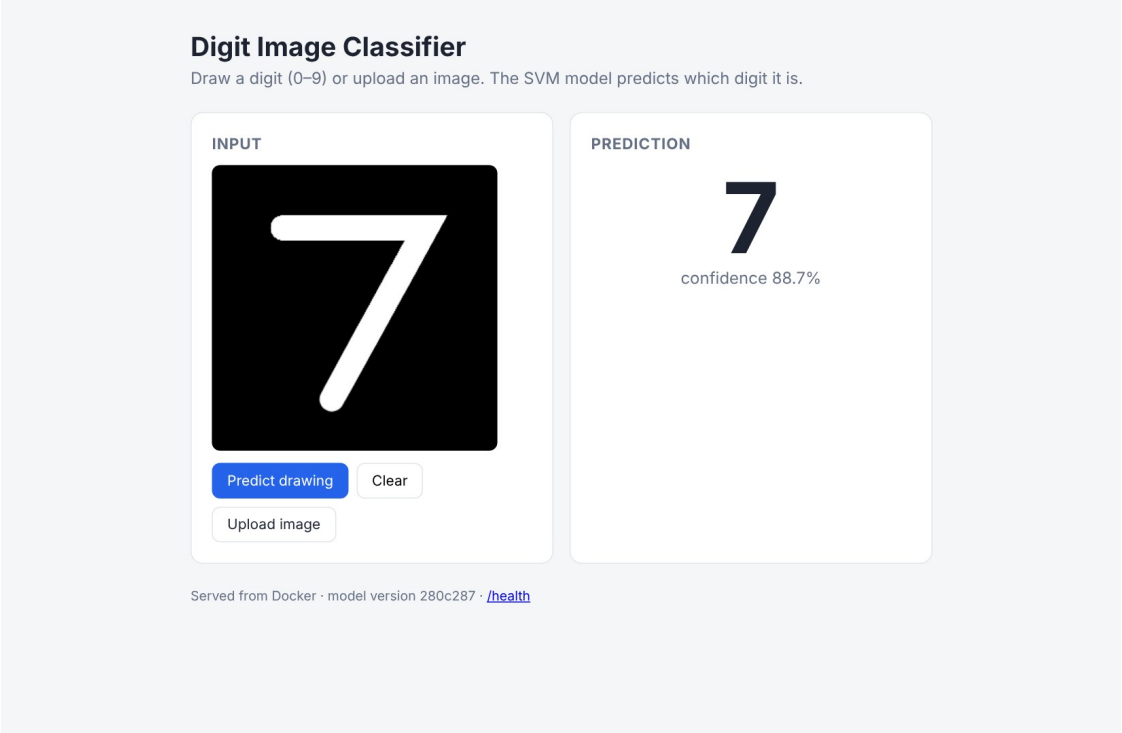
[INSERT SCREENSHOT] SSH session on EC2: `docker ps` showing `digit-classifier` on `0.0.0.0:80->5000/tcp`

9. Step 8: Verification and application execution

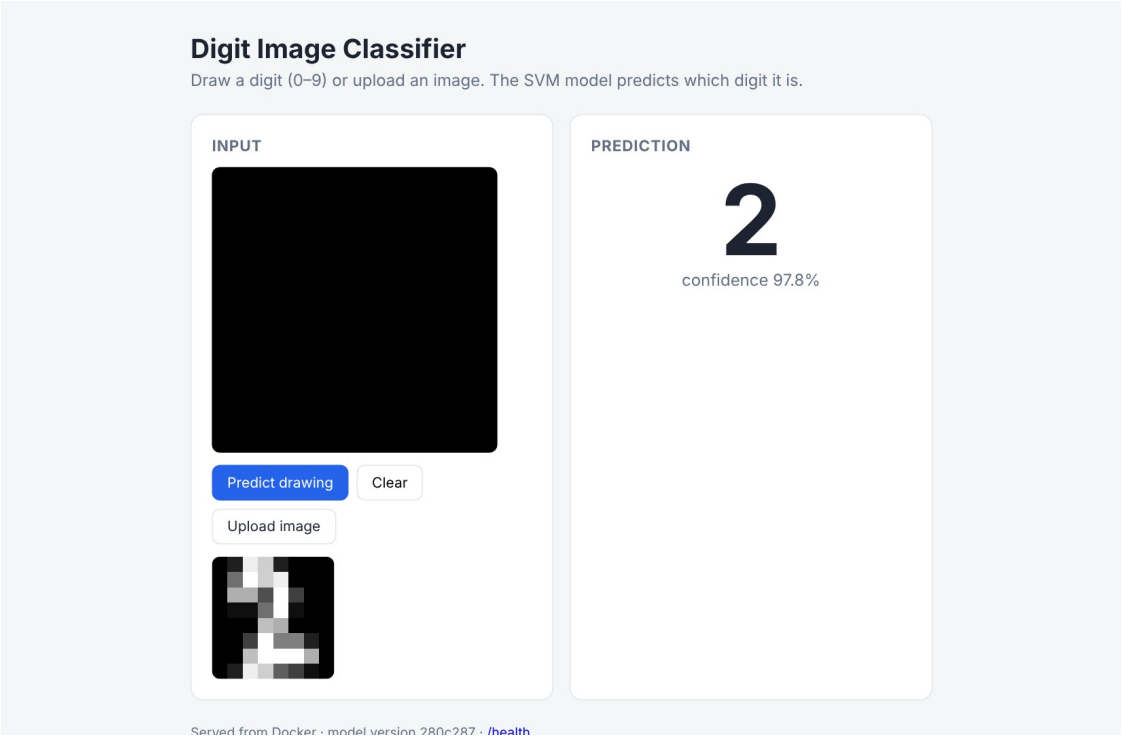
From any machine, the application is reachable on the instance's public IP:

```
curl http://<EC2-IP>/health                                    # {"status":"ok"}
curl -F image=@sample_images/digit_2.png http://<EC2-IP>/predict  #
{"confidence":0.98,"label":2}
```

The screenshots below show the same Docker image running and classifying a hand-drawn digit and an uploaded image:



Web UI: hand-drawn "7" classified as 7



Web UI: uploaded image classified as 2

[INSERT SCREENSHOT] Browser at `http://<EC2-public-IP>/` showing a prediction

[INSERT SCREENSHOT] `curl http://<EC2-public-IP>/health` from your machine

Public application URL: `http://<EC2-public-IP>/`

10. Challenges and solutions

Challenge	Solution
The trained model must reach the Docker image without committing binaries to Git.	The model is a DVC output. In CI it is produced by `dvc repro` (or pulled from S3), uploaded as an artifact by the `test` job and downloaded by the `build` job before `docker build`.
CI must not fail before AWS is set up. Without S3 credentials, `dvc pull` would fail.	The workflow checks `secrets.AWS_ACCESS_KEY_ID`. With no key, it regenerates the dataset and reproduces the pipeline. The deploy job runs only when `vars.EC2_HOST` is set.
GHCR rejects image names with upper-case letters, and the GitHub username is `SidhantN12`.	The image name is lower-cased in the workflow: `ghcr.io/${GITHUB_REPOSITORY,,}`.
The EC2 instance cannot pull a private GHCR image.	The deploy step pipes the job's `GITHUB_TOKEN` into `docker login ghcr.io` on the instance, pulls, then logs out. No long-lived registry credential is stored on the server.
`pip install` failed inside `docker build` with `CERTIFICATE_VERIFY_FAILED` in the development environment, which sits behind a TLS-inspecting proxy.	An optional BuildKit secret (`--secret id=pip_ca,...`) supplies the proxy CA during the build only. It is not stored in the image, and normal builds (GitHub Actions, EC2) skip it.
The scikit-learn 1.9 deprecation of `SVC(probability=True)` (scheduled for removal in 1.11) would eventually break CI.	Switched to `CalibratedClassifierCV(SVC(...), ensemble=False)`, the recommended replacement.
User drawings do not look like the training data: dark ink on a light background, any size.	`preprocess()` inverts light-background images and resizes to 8×8. Tests cover both cases.
S3 had to be tested before AWS credentials were available.	Ran the `aws s3` and `dvc push/pull` commands against moto, a local S3-compatible emulator (Section 4.3). Only the endpoint setting differs, so the same commands work on real S3.
Deployment must be verifiable automatically.	A Docker `HEALTHCHECK`, a CI smoke test that sends a real prediction, and a post-deploy `curl /health` against the public IP.

11. Conclusion

The project implements a reproducible path from data to deployment:

- DVC versions the 1,797-image dataset and the model and stores them in S3. Every commit pins exact data and model hashes.
- GitHub Actions reproduces the pipeline, runs 12 tests, then builds, smoke-tests and publishes a Docker image.
- The image runs on AWS EC2 and is reachable on the instance's public IP.

A push to `main` carries a change through testing to production without manual steps.

12. Links

- **GitHub repository:** https://github.com/SidhantN12/image-classifier-ci
- **CI/CD run #1:** https://github.com/SidhantN12/image-classifier-ci/actions/runs/40217311681
- **Docker image:** `ghcr.io/sidhantn12/image-classifier-ci:latest`
- **Deployed application:** `http://<EC2-public-IP>/`